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Eyewear with direction of sound arrival detection

Abstract

Eyewear providing a visual indicator to a hearing-impaired user indicating a direction of arrival of a sound relative to the eyewear to help the user obtain greater awareness of the surrounding environment. An eyewear optical assembly includes an image display displaying the visual indicator discernable by the user and corresponding to the direction of arrival of the sound, even when a sound source is not viewable through the optical assembly. The image display also displays an image indicative of the sound source. A front portion of an eyewear frame includes a light array, where one or more lights of the light array is illuminated to indicate the direction of arrival and intensity of the sound source. An array of vibrating devices may also be used to indicate the direction of arrival and intensity of the sound source.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0324275	12/2018	Nishizawa	N/A	G06F 9/3004
2020/0271960	12/2019	Baek et al.	N/A	N/A
2023/0049262	12/2022	Ruttenberg	N/A	A61B 5/165
2024/0110807	12/2023	Pomes	N/A	G01C 21/3632

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
212785854	12/2020	CN	N/A
2014130893	12/2013	KR	N/A
2016066896	12/2015	KR	A61F 11/04
2019052994	12/2018	KR	A61F 11/045
1020190068308	12/2018	KR	N/A
102000282	12/2018	KR	N/A
2009156145	12/2008	WO	N/A
2021136962	12/2020	WO	N/A

OTHER PUBLICATIONS

Search Report and Written Opinion received in International Application No. PCT/US2022/049651 dated Mar. 13, 2023, 11 pages. cited by applicant

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Background/Summary

TECHNICAL FIELD
(1) The present subject matter relates to an eyewear device, e.g., smart glasses.
BACKGROUND
(2) Portable eyewear devices, such as smart glasses, headwear, and headgear available today integrate microphones and see-through displays. Users with less than perfect hearing may have issues identifying the source of sounds in their environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The drawing figures depict one or more implementations, by way of example only, not by way of limitations. In the figures, like reference numerals refer to the same or similar elements.
(2) FIGS. **1**A and **1**B are side views of an example hardware configuration of an eyewear device, which shows optical assemblies with an image display, and field of view adjustments are applied to a user interface presented on the image display based on detected head or eye movement by a user;
(3) FIG. **1**C is a front view of the eyewear device depicting a forward-facing direction;
(4) FIG. **1**D is a top cross-sectional view of a temple of the eyewear device of FIG. **1**A depicting a visible light camera, a head movement tracker for tracking the head movement of the user of the eyewear device, a vibration device, and a circuit board;
(5) FIG. **2**A is a rear view of an example hardware configuration of an eyewear device, which includes an eye scanner on a frame, for use in a system for identifying a user of the eyewear device;
(6) FIG. **2**B is a rear view of an example hardware configuration of another eyewear device, which includes an array of lights for indicating a direction and intensity of a sound;
(7) FIG. **2**C is a rear view of an example hardware configuration of another eyewear device, which includes an array of vibration devices for indicating a direction and intensity of a sound;
(8) FIGS. **2**D and **2**E are rear views of example hardware configurations of the eyewear device, including two different types of image displays;
(9) FIG. **3** shows a rear perspective view of the eyewear device of FIG. **2**A depicting an infrared emitter, an infrared camera, a frame front, a frame back, and a circuit board;
(10) FIG. **4** is a cross-sectional view taken through the infrared emitter and the frame of the eyewear device of FIG. **3**;
(11) FIG. **5** illustrates detecting eye gaze direction;
(12) FIG. **6** illustrates detecting eye position;
(13) FIG. **7** depicts an example of visible light captured by the left visible light camera as a left raw image and visible light captured by the right visible light camera as a right raw image;
(14) FIG. **8**A illustrates a light in the left half of the light array illuminated to indicate a direction and intensity of as sound;
(15) FIG. **8**B illustrates the light in the right half of the light array illuminated to indicate a direction and intensity of a louder sound;
(16) FIG. **8**C illustrates a left vibration device of an array of vibration devices vibrating to indicate a direction and intensity of a sound source;
(17) FIG. **8**D illustrates a right vibration device vibrating to indicate a direction and intensity of a louder sound source;
(18) FIG. **8**E illustrates an optical display used to indicate a direction and source of a sound;
(19) FIG. **8**F illustrates an optical display used to indicate a direction and source of a sound by the additional use of a camera;
(20) FIG. **9** depicts a high-level functional block diagram including example electronic components disposed in eyewear;

(21) FIG. **10** illustrates a method of operating the eyewear; and
(22) FIG. **11** illustrates a second method of operating the eyewear.

DETAILED DESCRIPTION

(23) Eyewear providing a visual indicator to a hearing-impaired user indicating a direction of arrival of a sound relative to the eyewear to help the user obtain greater awareness of the surrounding environment. An eyewear optical assembly includes an image display displaying the visual indicator discernable by the user and corresponding to the direction of arrival of the sound, even when a sound source is not viewable through the optical assembly. The image display also displays an image indicative of the sound source. A front portion of an eyewear frame includes a light array, where one or more lights of the light array is illuminated to indicate the direction of arrival and intensity of the sound source. An array of vibrating devices may also be used to indicate the direction of arrival and intensity of the sound source.

(24) Additional objects, advantages and novel features of the examples will be set forth in part in the description which follows, and in part will become apparent to those skilled in the art upon examination of the following and the accompanying drawings or may be learned by production or operation of the examples. The objects and advantages of the present subject matter may be realized and attained by means of the methodologies, instrumentalities and combinations particularly pointed out in the appended claims.

(25) In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent to those skilled in the art that the present teachings may be practiced without such details. In other instances, well known methods, procedures, components, and circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.

(26) The term “coupled” as used herein refers to any logical, optical, physical or electrical connection, link or the like by which signals or light produced or supplied by one system element are imparted to another coupled element. Unless described otherwise, coupled elements or devices are not necessarily directly connected to one another and may be separated by intermediate components, elements or communication media that may modify, manipulate or carry the light or signals.

(27) The orientations of the eyewear device, associated components and any complete devices incorporating an eye scanner and camera such as shown in any of the drawings, are given by way of example only, for illustration and discussion purposes. In operation for a particular variable optical processing application, the eyewear device may be oriented in any other direction suitable to the particular application of the eyewear device, for example up, down, sideways, or any other orientation. Also, to the extent used herein, any directional term, such as front, rear, inwards, outwards, towards, left, right, lateral, longitudinal, up, down, upper, lower, top, bottom and side, are used by way of example only, and are not limiting as to direction or orientation of any optic or component of an optic constructed as otherwise described herein.

(28) Reference now is made in detail to the examples illustrated in the accompanying drawings and discussed below.

(29) FIG. **1**A-B are side views of an example hardware configuration of an eyewear device **100**, which includes a right optical assembly **180**B with an image display **180**D (FIG. **2**A). Eyewear device **100** includes multiple visible light cameras **114**A-B (FIG. **7**) that form a stereo camera, of which the right visible light camera **114**B is located on a right temple **110**B. Eyewear device **100** includes microphones **145**A-D that form a microphone array, where the right front microphone **145**C is located on the right temple **110**B and the right back **145**D microphone is located on the right temple **125**B. To determine a direction of arrival of sounds from a sound source **152** in an environment, the microphones **145**A-D are coupled to a processor **932** (FIG. **9**) for digital processing of the signals from the microphone array. The microphones **145**A-D send data signals to

processor **932** that then determines a direction and intensity of the sound from sound source **152** (FIG. **8**A) relative to eyewear **100**. The processor **932** responsively controls an array of lights **260** (FIG. **2**B), an image display **180**C-D, and/or an array of vibration devices **270** (FIG. **2**C) in the eyewear **100** to indicate to the user the direction and intensity of the sound from the sound source **152** relative to eyewear **100**, as will be described in more detail with reference to FIG. **2**B, FIG. **2**C, FIG. **8**A-**8**F.

(30) FIG. **1**C illustrates a front view of the eyewear device **100**. The left and right visible light cameras **114**A-B have an image sensor that is sensitive to the visible light range wavelength. Each of the visible light cameras **114**A-B have a different frontward facing angle of coverage, for example, visible light camera **114**B has the depicted angle of coverage **111**B (FIG. **7**). The angle of coverage is an angle range which the image sensor of the visible light camera **114**A-B picks up electromagnetic radiation and generates images. Examples of such visible lights camera **114**A-B include a high-resolution complementary metal-oxide-semiconductor (CMOS) image sensor and a video graphic array (VGA) camera, such as 640p (e.g., 640×480 pixels for a total of 0.3 megapixels), 720p, or 1080p. Image sensor data from the visible light cameras **114**A-B are captured along with geolocation data, digitized by an image processor, and stored in a memory.

(31) To provide stereoscopic vision, visible light cameras **114**A-B may be coupled to an image processor (element **912** of FIG. **9**) for digital processing along with a timestamp in which the image of the scene is captured. Image processor **912** includes circuitry to receive signals from the visible light camera **114**A-B and process those signals from the visible light cameras **114**A-B into a format suitable for storage in the memory (element **934** of FIG. **9**). The timestamp can be added by the image processor **912** or other processor, which controls operation of the visible light cameras **114**A-B. Visible light cameras **114**A-B allow the stereo camera to simulate human binocular vision. Stereo cameras provide the ability to reproduce three-dimensional images (element **715** of FIG. **7**) based on two captured images (elements **758**A-B of FIG. **7**) from the visible light cameras **114**A-B, respectively, having the same timestamp. Such three-dimensional images **715** allow for an immersive life-like experience, e.g., for virtual reality or video gaming. For stereoscopic vision, the pair of images **758**A-B are generated at a given moment in time—one image for each of the left and right visible light cameras **114**A-B. When the pair of generated images **758**A-B from the frontward facing angles of coverage **111**A-B of the left and right visible light cameras **114**A-B are stitched together (e.g., by the image processor **912**), depth perception is provided by the optical assembly **180**A-B.

(32) In an example, a user interface field of view adjustment system includes the eyewear device **100**. The eyewear device **100** includes a frame **105**, a right temple **110**B extending from a right lateral side **170**B of the frame **105**, and a see-through image display **180**D (FIGS. **2**A-B) comprising optical assembly **180**B to present a graphical user interface to a user. The eyewear device **100** includes the left visible light camera **114**A connected to the frame **105** or the left temple **110**A to capture a first image of the scene. Eyewear device **100** further includes the right visible light camera **114**B connected to the frame **105** or the right temple **110**B to capture (e.g., simultaneously with the left visible light camera **114**A) a second image of the scene which partially overlaps the first image. Although not shown in FIGS. **1**A-D, the user interface field of view adjustment system further includes a processor **932** (FIG. **9**) coupled to the eyewear device **100** and connected to the visible light cameras **114**A-B, the memory **934** accessible to the processor **932**, and programming in the memory **934**, for example in the eyewear device **100** itself or another part of the user interface field of view adjustment system.

(33) Although not shown in FIG. **1**A, the eyewear device **100** also includes a head movement tracker (element **109** of FIG. **1**D) or an eye movement tracker (element **213** of FIG. **2**B). Eyewear device **100** further includes the see-through image displays **180**C-D of optical assembly **180**A-B for presenting a sequence of displayed images, and an image display driver (element **942** of FIG. **9**) coupled to the see-through image displays **180**C-D of optical assembly **180**A-B to control the

image displays **180**C-D of optical assembly **180**A-B to present the sequence of displayed images **715**, which are described in further detail below. Eyewear device **100** further includes the memory **934** and the processor **932** having access to the image display driver **942** and the memory **934**. Eyewear device **100** further includes programming (element **934** of FIG. **9**) in the memory. Execution of the programming by the processor **932** configures the eyewear device **100** to perform functions, including functions to present, via the see-through image displays **180**C-D, an initial displayed image of the sequence of displayed images, the initial displayed image having an initial field of view corresponding to an initial head direction or an initial eye gaze direction (element **230** of FIG. **5**).

(34) Execution of the programming by the processor **932** further configures the eyewear device **100** to detect movement of a user of the eyewear device by: (i) tracking, via the head movement tracker (element **109** of FIG. **1**D), a head movement of a head of the user, or (ii) tracking, via an eye movement tracker (element **213** of FIG. **2**B, FIG. **5**), an eye movement of an eye of the user of the eyewear device **100**. Execution of the programming by the processor **932** further configures the eyewear device **100** to determine a field of view adjustment to the initial field of view of the initial displayed image based on the detected movement of the user. The field of view adjustment includes a successive field of view corresponding to a successive head direction or a successive eye direction. Execution of the programming by the processor **932** further configures the eyewear device **100** to generate a successive displayed image of the sequence of displayed images based on the field of view adjustment. Execution of the programming by the processor **932** further configures the eyewear device **100** to present, via the see-through image displays **180**C-D of the optical assembly **180**A-B, the successive displayed images.

(35) FIG. **1**D is a top cross-sectional view of the temple of the eyewear device **100** of FIG. **1**A depicting the right visible light camera **114**B, a head movement tracker **109**, and a circuit board **140**. Construction and placement of the left visible light camera **114**A is substantially similar to the right visible light camera **114**B, except the connections and coupling are on the left lateral side **170**A. As shown, the eyewear device **100** includes the right visible light camera **114**B and the circuit board **140**, which may be a flexible printed circuit board (PCB). The right hinge **226**B connects the right temple **110**B to a right temple **125**B of the eyewear device **100**. In some examples, components of the right visible light camera **114**B, the flexible PCB **140**, or other electrical connectors or contacts may be located on the right temple **125**B or the right hinge **226**B.

(36) As shown, eyewear device **100** has a head movement tracker **109**, which includes, for example, an inertial measurement unit (IMU). An inertial measurement unit is an electronic device that measures and reports a body's specific force, angular rate, and sometimes the magnetic field surrounding the body, using a combination of accelerometers and gyroscopes, sometimes also magnetometers. The inertial measurement unit works by detecting linear acceleration using one or more accelerometers and rotational rate using one or more gyroscopes. Typical configurations of inertial measurement units contain one accelerometer, gyro, and magnetometer per axis for each of the three axes: horizontal axis for left-right movement (X), vertical axis (Y) for top-bottom movement, and depth or distance axis for up-down movement (Z). The accelerometer detects the gravity vector. The magnetometer defines the rotation in the magnetic field (e.g., facing south, north, etc.) like a compass which generates a heading reference. The three accelerometers to detect acceleration along the horizontal, vertical, and depth axis defined above, which can be defined relative to the ground, the eyewear device **100**, or the user wearing the eyewear device **100**.

(37) Eyewear device **100** detects movement of the user of the eyewear device **100** by tracking, via the head movement tracker **109**, the head movement of the head of the user. The head movement includes a variation of head direction on a horizontal axis, a vertical axis, or a combination thereof from the initial head direction during presentation of the initial displayed image on the image display. In one example, tracking, via the head movement tracker **109**, the head movement of the head of the user includes measuring, via the inertial measurement unit **109**, the initial head

direction on the horizontal axis (e.g., X axis), the vertical axis (e.g., Y axis), or the combination thereof (e.g., transverse or diagonal movement). Tracking, via the head movement tracker **109**, the head movement of the head of the user further includes measuring, via the inertial measurement unit **109**, a successive head direction on the horizontal axis, the vertical axis, or the combination thereof during presentation of the initial displayed image.

(38) Tracking, via the head movement tracker **109**, the head movement of the head of the user further includes determining the variation of head direction based on both the initial head direction and the successive head direction. Detecting movement of the user of the eyewear device **100** further includes in response to tracking, via the head movement tracker **109**, the head movement of the head of the user, determining that the variation of head direction exceeds a deviation angle threshold on the horizontal axis, the vertical axis, or the combination thereof. The deviation angle threshold is between about 3° to 10°. As used herein, the term "about" when referring to an angle means±10% from the stated amount.

(39) Variation along the horizontal axis slides three-dimensional objects, such as characters, bitmojis, application icons, etc. in and out of the field of view by, for example, hiding, unhiding, or otherwise adjusting visibility of the three-dimensional object. Variation along the vertical axis, for example, when the user looks upwards, in one example, displays weather information, time of day, date, calendar appointments, etc. In another example, when the user looks downwards on the vertical axis, the eyewear device **100** may power down.

(40) The right temple **110**B includes temple body **211** and a temple cap, with the temple cap omitted in the cross-section of FIG. **1**D. Disposed inside the right temple **110**B are various interconnected circuit boards, such as flexible PCB **140**, that include controller circuits for right visible light camera **114**B, inner microphone(s) **130**, speaker(s) **132**, low-power wireless circuitry (e.g., for wireless short-range network communication via Bluetooth™), high-speed wireless circuitry (e.g., for wireless local area network communication via WiFi).

(41) The right visible light camera **114**B is coupled to or disposed on the flexible PCB **140** and covered by a visible light camera cover lens, which is aimed through opening(s) formed in the right temple **110**B. In some examples, the frame **105** connected to the right temple **110**B includes the opening(s) for the visible light camera cover lens. The frame **105** includes a front-facing side configured to face outwards away from the eye of the user. The opening for the visible light camera cover lens is formed on and through the front-facing side. In the example, the right visible light camera **114**B has an outward facing angle of coverage **111**B with a line of sight or perspective of the right eye of the user of the eyewear device **100**. The visible light camera cover lens can also be adhered to an outward facing surface of the right temple **110**B in which an opening is formed with an outward facing angle of coverage, but in a different outwards direction. The coupling can also be indirect via intervening components.

(42) Left (first) visible light camera **114**A is connected to the left see-through image display **180**C of left optical assembly **180**A to generate a first background scene of a first successive displayed image. The right (second) visible light camera **114**B is connected to the right see-through image display **180**D of right optical assembly **180**B to generate a second background scene of a second successive displayed image. The first background scene and the second background scene partially overlap to present a three-dimensional observable area of the successive displayed image.

(43) The PCB **140** is also electrically coupled to the array of lights **260** (FIG. **2**B) including lights **262** located in the frame **105**, such as bridge **106**, that selectively illuminate to indicate a direction of arrival of sound from the sound source **152** in reference to an orientation of the eyewear **100**, as determined using the microphone array **145**A-D. The light **262** in the light array **260** that is illuminated indicates the direction of arrival of the sound from the sound source **152** (FIG. **8**A-**8**B), and the relative brightness of the illuminated light **262** indicates the intensity of the sound from sound source **152** in reference to the surrounding environmental noise, where a relatively brighter light indicates a louder sound. The PCB **140** is also electrically coupled to haptics (FIG. **2**C), such

as the array of vibration devices **270** including vibration devices **272** located in the frame **105**, such as in the bridge **106**, which selectively vibrate to indicate an arrival direction of sound from the sound source **152** in reference to the orientation of the eyewear **100** as determined using the microphone array **145**A-D. The vibration device **272** in the vibration device array **270** that is energized (FIG. **8**C-**8**D) indicates an arrival direction of sound from the sound source **152**, and the relative strength of vibration of the vibration device **272** indicates the intensity of sound from the sound source **152**, where a relatively stronger vibration indicates a louder sound. Similarly, a flexible PCB **140** may be disposed inside the left temple **110**A and is coupled to one or more other components housed in the left temple **110**A.

(44) FIG. **2**A is a rear view of an example hardware configuration of an eyewear device **100**, which includes an eye scanner **113** on a frame **105**, for use in a system for determining an eye position and gaze direction of a wearer/user of the eyewear device **100**. As shown in FIG. **2**A, the eyewear device **100** is in a form configured for wearing by a user, which are eyeglasses in the example of FIG. **2**A. The eyewear device **100** can take other forms and may incorporate other types of frameworks, for example, a headgear, a headset, or a helmet.

(45) In the eyeglasses example, eyewear device **100** includes the frame **105** which includes the left rim **107**A connected to the right rim **107**B via the bridge **106** adapted for a nose of the user. The left and right rims **107**A-B include respective apertures **175**A-B which hold the respective optical element **180**A-B, such as a lens and the see-through displays **180**C-D. As used herein, the term lens is meant to cover transparent or translucent pieces of glass or plastic having curved and flat surfaces that cause light to converge/diverge or that cause little or no convergence/divergence.

(46) Although shown as having two optical elements **180**A-B, the eyewear device **100** can include other arrangements, such as a single optical element depending on the application or intended user of the eyewear device **100**. As further shown, eyewear device **100** includes the left temple **110**A adjacent the left lateral side **170**A of the frame **105** and the right temple **110**B adjacent the right lateral side **170**B of the frame **105**. The temples **110**A-B may be integrated into the frame **105** on the respective sides **170**A-B (as illustrated) or implemented as separate components attached to the frame **105** on the respective sides **170**A-B. Alternatively, the temples **110**A-B may be integrated into temples (not shown) attached to the frame **105**.

(47) In the example of FIG. **2**A, the eye scanner **113** includes an infrared emitter **115** and an infrared camera **120**. Visible light cameras typically include a blue light filter to block infrared light detection, in an example, the infrared camera **120** is a visible light camera, such as a low-resolution video graphic array (VGA) camera (e.g., 640×480 pixels for a total of 0.3 megapixels), with the blue filter removed. The infrared emitter **115** and the infrared camera **120** are co-located on the frame **105**, for example, both are shown as connected to the upper portion of the left rim **107**A. The frame **105** or one or more of the left and right temples **110**A-B include a circuit board (not shown) that includes the infrared emitter **115** and the infrared camera **120**. The infrared emitter **115** and the infrared camera **120** can be connected to the circuit board by soldering, for example.

(48) Other arrangements of the infrared emitter **115** and infrared camera **120** can be implemented, including arrangements in which the infrared emitter **115** and infrared camera **120** are both on the right rim **107**B, or in different locations on the frame **105**, for example, the infrared emitter **115** is on the left rim **107**A and the infrared camera **120** is on the right rim **107**B. In another example, the infrared emitter **115** is on the frame **105** and the infrared camera **120** is on one of the temples **110**A-B, or vice versa. The infrared emitter **115** can be connected essentially anywhere on the frame **105**, left temple **110**A, or right temple **110**B to emit a pattern of infrared light. Similarly, the infrared camera **120** can be connected essentially anywhere on the frame **105**, left temple **110**A, or right temple **110**B to capture at least one reflection variation in the emitted pattern of infrared light.

(49) The infrared emitter **115** and infrared camera **120** are arranged to face inwards towards an eye of the user with a partial or full field of view of the eye in order to identify the respective eye position and gaze direction. For example, the infrared emitter **115** and infrared camera **120** are

positioned directly in front of the eye, in the upper part of the frame **105** or in the temples **110**A-B at either ends of the frame **105**.
(50) FIG. **2**B is a rear view of an example hardware configuration of another eyewear device **200**. In this example configuration, the eyewear device **200** is depicted as including an eye scanner **213** on a right temple **210**B. As shown, an infrared emitter **215** and an infrared camera **220** are co-located on the right temple **210**B. It should be understood that the eye scanner **213** or one or more components of the eye scanner **213** can be located on the left temple **210**A and other locations of the eyewear device **200**, for example, the frame **105**. The infrared emitter **215** and infrared camera **220** are like that of FIG. **2**A, but the eye scanner **213** can be varied to be sensitive to different light wavelengths as described previously in FIG. **2**A. Similar to FIG. **2**A, the eyewear device **200** includes a frame **105** which includes a left rim **107**A which is connected to a right rim **107**B via a bridge **106**. The left and right rims **107**A-B include respective apertures which hold the respective optical elements **180**A-B comprising the see-through display **180**C-D.
(51) Also shown in FIG. **2**B is the generally linear light array **260** including the plurality of lights **262**, such as light emitting diodes (LEDs), formed in the frame **105** above each of the optical elements **180**A-B and facing rearwardly. The processor **932** selectively controls the individual lights **262** (FIG. **9**) of the light array **260** to indicate to a partially deaf or deaf individual the arrival direction and intensity of sound from the sound source **152** relative to a true direction forward. In the example shown in FIG. **8**A, the fifth light **262** left of the center light shown illuminated in the light array **260** corresponds to the arrival direction of sound from the sound source **152**, and it may indicate the sound source **152** is angle A degrees left of center, such as 35 degrees, and the relative brightness of the light **262** indicates intensity of the sound from the sound source **152**, such as 90 decibels.
(52) Shown in FIG. **2**C is the array of vibration devices **270** including the plurality of vibration devices **272** formed in the frame **105** above each of the optical elements **180**A-B and facing rearwardly. The processor **932** selectively controls the individual vibration devices **272** (FIG. **9**) of the vibration device array **270** to indicate to a partially deaf or deaf individual the arrival direction and intensity of the sound from the sound source **152** relative to a true direction forward. In the example shown in FIG. **8**C, the fifth vibration device **262** left of the center light shown vibrating corresponds to the arrival direction of sound from the sound source **152** and may indicate the sound source is angle A degrees left of center, such as 35 degrees, and the relative vibration of the vibration device **272** indicates the intensity of the sound from the sound source **152** from the eyewear **100**, such as 90 decibels.
(53) FIGS. **2**D-E are rear views of example hardware configurations of the eyewear device **100**, including two different types of see-through image displays **180**C-D. In one example, these see-through image displays **180**C-D of optical assembly **180**A-B include an integrated image display. As shown in FIG. **2**D, the optical assemblies **180**A-B includes a suitable display matrix **180**C-D of any suitable type, such as a liquid crystal display (LCD), an organic light-emitting diode (OLED) display, a waveguide display, or any other such display. The optical assembly **180**A-B also includes an optical layer or layers **176**, which can include lenses, optical coatings, prisms, mirrors, waveguides, optical strips, and other optical components in any combination. The optical layers **176**A-N can include a prism having a suitable size and configuration and including a first surface for receiving light from display matrix and a second surface for emitting light to the eye of the user. The prism of the optical layers **176**A-N extends over all or at least a portion of the respective apertures **175**A-B formed in the left and right rims **107**A-B to permit the user to see the second surface of the prism when the eye of the user is viewing through the corresponding left and right rims **107**A-B. The first surface of the prism of the optical layers **176**A-N faces upwardly from the frame **105** and the display matrix overlies the prism so that photons and light emitted by the display matrix impinge the first surface. The prism is sized and shaped so that the light is refracted within the prism and is directed towards the eye of the user by the second surface of the prism of the

optical layers **176**A-N. In this regard, the second surface of the prism of the optical layers **176**A-N can be convex to direct the light towards the center of the eye. The prism can optionally be sized and shaped to magnify the image projected by the see-through image displays **180**C-D, and the light travels through the prism so that the image viewed from the second surface is larger in one or more dimensions than the image emitted from the see-through image displays **180**C-D.

(54) In another example, the see-through image displays **180**C-D of optical assembly **180**A-B include a projection image display as shown in FIG. **2**E. The optical assembly **180**A-B includes a laser projector **150**, which is a three-color laser projector using a scanning mirror or galvanometer. During operation, an optical source such as a laser projector **150** is disposed in or on one of the temples **125**A-B of the eyewear device **100**. Optical assembly **180**A-B includes one or more optical strips **155**A-N spaced apart across the width of the lens of the optical assembly **180**A-B or across a depth of the lens between the front surface and the rear surface of the lens.

(55) As the photons projected by the laser projector **150** travel across the lens of the optical assembly **180**A-B, the photons encounter the optical strips **155**A-N. When a particular photon encounters a particular optical strip, the photon is either redirected towards the user's eye, or it passes to the next optical strip. A combination of modulation of laser projector **150**, and modulation of optical strips, may control specific photons or beams of light. In an example, a processor controls optical strips **155**A-N by initiating mechanical, acoustic, or electromagnetic signals. Although shown as having two optical assemblies **180**A-B, the eyewear device **100** can include other arrangements, such as a single or three optical assemblies, or the optical assembly **180**A-B may have arranged different arrangement depending on the application or intended user of the eyewear device **100**.

(56) As further shown in FIGS. **2**D-E, eyewear device **100** includes a left temple **110**A adjacent the left lateral side **170**A of the frame **105** and a right temple **110**B adjacent the right lateral side **170**B of the frame **105**. The temples **110**A-B may be integrated into the frame **105** on the respective lateral sides **170**A-B (as illustrated) or implemented as separate components attached to the frame **105** on the respective sides **170**A-B. Alternatively, the temples **110**A-B may be integrated into temples **125**A-B attached to the frame **105**.

(57) In one example, the see-through image displays include the first see-through image display **180**C and the second see-through image display **180**D. Eyewear device **100** includes first and second apertures **175**A-B which hold the respective first and second optical assembly **180**A-B. The first optical assembly **180**A includes the first see-through image display **180**C (e.g., a display matrix of FIG. **2**C or optical strips **155**A-N′ and a projector **150**A). The second optical assembly **180**B includes the second see-through image display **180**D e.g., a display matrix of FIG. **2**D or optical strips **155**A-N″ and a projector **150**B. The successive field of view of the successive displayed image includes an angle of view between about 15° to 30, and more specifically 24°, measured horizontally, vertically, or diagonally. The successive displayed image having the successive field of view represents a combined three-dimensional observable area visible through stitching together of two displayed images presented on the first and second image displays.

(58) As used herein, “an angle of view” describes the angular extent of the field of view associated with the displayed images presented on each of the left and right image displays **180**C-D of optical assembly **180**A-B. The “angle of coverage” describes the angle range that a lens of visible light cameras **114**A-B or infrared camera **220** can image. Typically, the image circle produced by a lens is large enough to cover the film or sensor completely, possibly including some vignetting (i.e., a reduction of an image's brightness or saturation toward the periphery compared to the image center). If the angle of coverage of the lens does not fill the sensor, the image circle will be visible, typically with strong vignetting toward the edge, and the effective angle of view will be limited to the angle of coverage. The “field of view” is intended to describe the field of observable area which the user of the eyewear device **100** can see through his or her eyes via the displayed images presented on the left and right image displays **180**C-D of the optical assembly **180**A-B. Image

display **180**C of optical assembly **180**A-B can have a field of view with an angle of coverage between 15° to 30°, for example 24°, and have a resolution of 480×480 pixels.

(59) FIG. **3** shows a rear cutaway perspective view of the eyewear device of FIG. **2**A. The eyewear device **100** includes an infrared emitter **215**, infrared camera **220**, a frame front **330**, a frame back **335**, and a circuit board **340**. It can be seen in FIG. **3** that the upper portion of the left rim of the frame of the eyewear device **100** includes the frame front **330** and the frame back **335**. An opening for the infrared emitter **215** is formed on the frame back **335**.

(60) As shown in the encircled cross-section **4** in the upper middle portion of the left rim of the frame, a circuit board, which is a flexible PCB **340**, is sandwiched between the frame front **330** and the frame back **335**. Also shown in further detail is the attachment of the left temple **110**A to the left temple **325**A via the left hinge **326**A. In some examples, components of the eye movement tracker **213**, including the infrared emitter **215**, the flexible PCB **340**, or other electrical connectors or contacts may be located on the left temple **325**A or the left hinge **326**A.

(61) FIG. **4** is a cross-sectional view through the infrared emitter **215** and the frame corresponding to the encircled cross-section **4** of the eyewear device of FIG. **3**. Multiple layers of the eyewear device **100** are illustrated in the cross-section of FIG. **4**, as shown the frame includes the frame front **330** and the frame back **335**. The flexible PCB **340** is disposed on the frame front **330** and connected to the frame back **335**. The infrared emitter **215** is disposed on the flexible PCB **340** and covered by an infrared emitter cover lens **445**. For example, the infrared emitter **215** is reflowed to the back of the flexible PCB **340**. Reflowing attaches the infrared emitter **215** to contact pad(s) formed on the back of the flexible PCB **340** by subjecting the flexible PCB **340** to controlled heat which melts a solder paste to connect the two components. In one example, reflowing is used to surface mount the infrared emitter **215** on the flexible PCB **340** and electrically connect the two components. However, it should be understood that through-holes can be used to connect leads from the infrared emitter **215** to the flexible PCB **340** via interconnects, for example.

(62) The frame back **335** includes an infrared emitter opening **450** for the infrared emitter cover lens **445**. The infrared emitter opening **450** is formed on a rear-facing side of the frame back **335** that is configured to face inwards towards the eye of the user. In the example, the flexible PCB **340** can be connected to the frame front **330** via the flexible PCB adhesive **460**. The infrared emitter cover lens **445** can be connected to the frame back **335** via infrared emitter cover lens adhesive **455**. The coupling can also be indirect via intervening components.

(63) In an example, the processor **932** utilizes eye tracker **213** to determine an eye gaze direction **230** of a wearer's eye **234** as shown in FIG. **5**, and an eye position **236** of the wearer's eye **234** within an eyebox as shown in FIG. **6**. The eye tracker **213** is a scanner which uses infrared light illumination (e.g., near-infrared, short-wavelength infrared, mid-wavelength infrared, long-wavelength infrared, or far infrared) to captured image of reflection variations of infrared light from the eye **234** to determine the gaze direction **230** of a pupil **232** of the eye **234**, and also the eye position **236** with respect to the see-through display **180**D.

(64) FIG. **7** depicts an example of capturing visible light with cameras. Visible light is captured by the left visible light camera **114**A with a left visible light camera field of view **111**A as a left raw image **758**A. Visible light is captured by the right visible light camera **114**B with a right visible light camera field of view **111**B as a right raw image **758**B. Based on processing of the left raw image **758**A and the right raw image **758**B, a three-dimensional depth map **715** of a three-dimensional scene, referred to hereafter as an image, is generated by processor **932**.

(65) FIG. **8**A and FIG. **8**B illustrate examples of the light array **260** positioned on frame **105**, including on the bridge **106**, for improving the user experience of users of eyewear **100**/**200** having partial deafness or complete deafness. To compensate for deafness, as described earlier with respect to FIG. **2**B, the lights **262** of light array **260** are selectively turned on. The processor **932** selectively controls the individual lights **262** of the light array **260** to indicate to a deaf individual the arrival direction and intensity of the sound from the sound source **152** relative to a true direction forward.

(66) In the example shown in FIG. **8**A, the fifth light **262** left of the center light shown illuminated in the light array **260** corresponds to the arrival direction of sound from the sound source **152**, and the light may indicate the sound source **152** is angle A degrees left of center, such as 35 degrees, and the relative brightness of the light **262** indicates the intensity of the sound from the sound source **152**, such as 90 decibels. In the example shown in FIG. **8**B, the fourth light **262** right of the center light shown illuminated in the light array **260** corresponds to the arrival direction of sound from the sound source **152**, and it may indicate the sound source **152** is angle A degrees right of center, such as 28 degrees, and the relative brightness of the light **262** indicates intensity of the sound from the sound source **152**, such as 70 decibels.
(67) In an example, each of the six lights **262** left and right of the center light **262** are uniformly spaced and indicate 7-degree increments. In an example, the first light **262** left of the center light is illuminated if the sound source **152** is 7 degrees left of true forward, the second light **262** left of center light **262** is illuminated if the sound source **152** is 14 degrees left of true forward, and so on. The lights on the far left or far right side from center may be illuminated to indicate a sound source **152** that is originating from out of view of the user.
(68) In an example, both the far left and far right lights may be illuminated to indicate a sound source **152** that is located directly behind the user. If the intensity of sound from the sound source **152** is less than 50 decibels, the corresponding light has a 20 percent intensity. The user may customize the minimum intensity threshold of the sound detected by the microphones **145** required to trigger the processor **934** to illuminate the light array **260**.
(69) FIG. **8**C and FIG. **8**D illustrate examples of the vibration device array **270** positioned on frame **105**, including on the bridge **106**, for improving the user experience of users of eyewear **100**/**200** having partial or complete deafness. To compensate for deafness, as described earlier with respect to FIG. **2**C, the vibration devices **272** of vibration device array **270** are selectively turned on. The processor **932** selectively controls the individual vibration devices **272** of the vibration device array **270** to indicate to a near to deaf or completely deaf individual the arrival direction and intensity of sound from the sound source **152** relative to a true direction forward.
(70) In the example shown in FIG. **8**C, the fifth vibration device **272** left of the center vibration device shown vibrating in the vibration device array **270** corresponds to a direction of the sound source **152**, and it may indicate the sound source **152** is angle A degrees left of center, such as 35 degrees, and the relative vibration intensity of the vibration device **272** indicates the intensity of the sound from the sound source **152**, such as 90 decibels. In the example shown in FIG. **8**D, the fourth vibration device **272** right of the center vibration device shown vibrating in the vibration device array **270** corresponds to the arrival direction of sound from the sound source **152**, and it may indicate the sound source **152** is angle A degrees right of center, such as 28 degrees, and the relative vibration of the vibration device **272** indicates intensity of the sound source **152**, such as 80 decibels.
(71) In an example, each of the six vibrations devices **272** left and right of the center vibration device **272** are uniformly spaced and indicate 7-degree increments. In an example, the first vibration device **272** left of the center vibration device is vibrated if the sound source **152** is 7 degrees left of true forward, the second vibration device **272** left of center is vibrated if the sound source is 14 degrees left of true forward, and so on.
(72) In an example, the vibration devices on the far left or far right side from center may be vibrated to indicate the sound source **152** that is originating from out of view of the user. In an example, both the far left and far right vibration devices may be vibrated to indicate a sound source **152** that is located directly behind the user.
(73) The relative vibration of the vibration devices **272** is controlled by processor **932** to correspond to an intensity of the sound from the sound source **152** as measured by the microphones **145**A-D. In an example, if the sound from the sound source **152** has an intensity above 100 decibels, the corresponding vibration device has a 100 percent intensity. If the sound from the

sound source **152** has an intensity between 100-80 decibels, the corresponding vibration device has a 75 percent intensity. If the sound intensity is less than 50 decibels, the corresponding vibration device has a 50 intensity. In another example, the vibration intensity of each vibration device **272** can also be controlled such that the vibration intensity varies directly with the intensity of the sound from sound source **152**.

(74) The processor **932** is also configured to identify the sound source **152** as will be described in more detail with reference to FIG. **9**. The processor **932** is configured to identify the sound source **152** and control the vibrational device array **270** to vibrate a specific pattern for a specified sound source. For example, a sound source identified by the processor **932** as an alarm may cause the vibrational array **270** to vibrate in a pattern **3** quick vibrations repeated for the duration of the alarm. The user may customize the minimum intensity threshold of the sound detected by the microphone array **145** required to trigger the processor **934** to activate the vibrational array **270**.

(75) FIG. **8**E illustrates an example of using the image display **180**C-D of the eyewear **100**/**200** to display a visual indicator **275** on display **180**C-D. The processor **932** selectively controls the indicator **275** displayed on the image display **180**C-D to indicate the arrival direction and intensity of sound from the sound source **152** relative to a true direction forward. In an example, the indicator **275** can be selectively configured to be a 3D arrow, a 2D arrow, a directional heading, shading of part of the image display **180**C-D to indicate direction, or custom type of indicator. The brightness, color, sharpness, line thickness, or line type of the indicator **275** can be selectively configured to indicate the intensity of the sound. The processor **932** is configured to identify the sound from the sound source **152** by comparing the signals from the microphones **145**A-D to a library of known sounds **946** by using a detection algorithm **945** as further described in FIG. **9**. The library **946** includes a plurality of sounds correlated to a plurality of sources. The identity of the sound source is then displayed as part of the indicator **275** as well as the arrival direction of sound from sound source **152**. The library **946** is stored in memory, such as in eyewear memory **934**, or on server **998** and accessed using network **995**.

(76) In an example, if a car is approaching from behind a deaf person wearing the eyewear **100**/**200**, the microphones **145**A-D detect the sound of the car engine from behind the user and send signals to the processor **932**. The processor **932** identifies the direction of arrival and the identity of the sound from the sound source **152** and sends a signal to the image display **180**D to display indicator **275** to inform the user that a car is approaching from behind the user. In an example, the indicator **275** consists of a 3D directional arrow indicating the direction the sound is arriving from, and an image indicating that the sound source is identified as a car, such as an image of a car next to the arrow. The indicator **275** is also configured to inform the user as to how loud the sound is. For example, the indicator **275** may flash rapidly for a sound with an intensity over 100 decibels, and it may have an intensity corresponding to the intensity of the sound for sounds under 100 decibels. In another example, the indicator **275** may consist of shading around the image display **180**C-D to indicate a looping, or ambient, noise such as a bus engine. The darkness of the shading can also be used to indicate the intensity of the noise. In another example, if a user is waiting at a bus stop and does not notice a bus pulling into the stop located behind and to the left of the user, the image display **180**C-D becomes shaded in the bottom left corner of the display to indicate a new ambient sound arriving from that direction to give the user awareness of their environment. Another example includes the presentation of a visual signature of the sound on the image display **180**C-D. Given the pattern-matching abilities of the user's brain, the user can learn to recognize contextually relevant sounds by their visual signature. For example, the image display **180**C-D displays a live feed of the visual sound signature of the environment while a user is sitting in their living room at home. If a visitor knocks on their front door the visual sound signature corresponding to the knock is displayed to the user. The hearing impaired user of the eyewear **100**/**200** then recognizes the visual sound signature as a knock on their door. The user may customize the minimum intensity of the sound detected by the microphones **145** required to trigger

the processor **934** to produce the indicator **275**. In another example, if a hearing-impaired user of the eyewear **100**/**200** is sitting in the living room of their home, the user can be alerted to a child going into a bathroom behind the user to draw a bath. The processor **932** identifies the sound of running water and displays an image or icon of a water tap as an indicator **275** on the image display **180**C-D to indicate the sound of running water behind user.

(77) FIG. **8**F illustrates an example of using a camera along with the microphones **145**A-D in determining the identity of the sound source **152**. In this embodiment, the cameras **114**A-B are computer vision (CV) cameras rather than visible light cameras. The processor **932** identifies the sound source by comparing the signals received by the microphones **145** to the sound library **946**, in combination with using the CV cameras performing object identification. In an example, the processor **932** uses the CV cameras **114**A-B to identify objects present in the environment surrounding the user of the eyewear **100**/**200** to assist in the identification of sounds received by the microphones **145**. In an example, the processor **932** identifies that there is an oven in the environment of the user of the eyewear **100**/**200** by using the CV cameras. When the oven timer goes off the microphones **145** detect the sound. The processor **932** then identifies the sound as the oven timer by comparing the received sound to the sound library **946**, and by correlating the sound using the CV camera object identification. The processor **932** then produces an indicator **275** on the image display **180**C-D to indicate that the oven timer is going off to the user, such as displaying an image or icon of an oven to the user. The indicator **275** may also indicate a direction of sound arrival relative to the user.

(78) FIG. **9** depicts a high-level functional block diagram including example electronic components disposed in eyewear **100** and **200**. Memory **934** includes instructions (in the form of code) for execution by processor **932** to implement functionality of eyewear **100**/**200**, including instructions for processor **932** to process signals from the array of microphones **145** and the CV cameras **114**A-B to determine direction of arrival and the identification of sound sources **152**. The instructions also control the lights **262** of light array **260**, the vibration devices **270**, and display the indicator **275** on the image display **180**C-D. The memory **934** of the eyewear **100**/**200** also contains the sound library **946** that includes known sounds and known sound producing objects that a user may encounter in their environment, although the sound library **946** may be stored in server **998** in another example. The processor **932** uses the detection algorithm **945** and the sound library **946** along with the array of microphones **145** to identify the arrival direction of sound, the intensity of the sound, and identification of the sound source. The processor **932** may also use the CV cameras for sound identification.

(79) A user interface adjustment system **900** includes a wearable device, which is the eyewear device **100** with an eye movement tracker **213** (e.g., shown as infrared emitter **215** and infrared camera **220** in FIG. **2**B). User interface adjustments system **900** also includes the mobile device **990** and a server system **998** connected via various networks. Mobile device **990** may be a smartphone, tablet, laptop computer, access point, or any other such device capable of connecting with eyewear device **100** using both a low-power wireless connection **925** and a high-speed wireless connection **937**. Mobile device **990** is connected to server system **998** and network **995**. The network **995** may include any combination of wired and wireless connections.

(80) Eyewear device **100** includes at least two visible light cameras **114**A-B (one associated with the left lateral side **170**A and one associated with the right lateral side **170**B). Eyewear device **100** further includes two see-through image displays **180**C-D of the optical assembly **180**A-B (one associated with the left lateral side **170**A and one associated with the right lateral side **170**B). The image displays **180**C-D are optional in this disclosure. Eyewear device **100** also includes image display driver **942**, image processor **912**, low-power circuitry **920**, and high-speed circuitry **930**. The components shown in FIG. **9** for the eyewear device **100** are located on one or more circuit boards, for example a PCB or flexible PCB, in the temples. Alternatively, or additionally, the depicted components can be located in the temples, frames, hinges, or bridge of the eyewear device

100. Left and right visible light cameras **114**A-B can include digital camera elements such as a complementary metal-oxide-semiconductor (CMOS) image sensor, charge coupled device, a lens, or any other respective visible or light capturing elements that may be used to capture data, including images of scenes with unknown objects.

(81) Eye movement tracking programming implements the user interface field of view adjustment instructions, including, to cause the eyewear device **100** to track, via the eye movement tracker **213**, the eye movement of the eye of the user of the eyewear device **100**. Other implemented instructions (functions) cause the eyewear device **100** to determine, a field of view adjustment to the initial field of view of an initial displayed image based on the detected eye movement of the user corresponding to a successive eye direction. Further implemented instructions generate a successive displayed image of the sequence of displayed images based on the field of view adjustment. The successive displayed image is produced as visible output to the user via the user interface. This visible output appears on the see-through image displays **180**C-D of optical assembly **180**A-B, which is driven by image display driver **934** to present the sequence of displayed images, including the initial displayed image with the initial field of view and the successive displayed image with the successive field of view.

(82) As shown in FIG. **9**, high-speed circuitry **930** includes high-speed processor **932**, memory **934**, and high-speed wireless circuitry **936**. In the example, the image display driver **942** is coupled to the high-speed circuitry **930** and operated by the high-speed processor **932** in order to drive the left and right image displays **180**C-D of the optical assembly **180**A-B. High-speed processor **932** may be any processor capable of managing high-speed communications and operation of any general computing system needed for eyewear device **100**. High-speed processor **932** includes processing resources needed for managing high-speed data transfers on high-speed wireless connection **937** to a wireless local area network (WLAN) using high-speed wireless circuitry **936**. In certain examples, the high-speed processor **932** executes an operating system such as a LINUX operating system or other such operating system of the eyewear device **100** and the operating system is stored in memory **934** for execution. In addition to any other responsibilities, the high-speed processor **932** executing a software architecture for the eyewear device **100** is used to manage data transfers with high-speed wireless circuitry **936**. In certain examples, high-speed wireless circuitry **936** is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as Wi-Fi. In other examples, other high-speed communications standards may be implemented by high-speed wireless circuitry **936**.

(83) Low-power wireless circuitry **924** and the high-speed wireless circuitry **936** of the eyewear device **100** can include short range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). Mobile device **990**, including the transceivers communicating via the low-power wireless connection **925** and high-speed wireless connection **937**, may be implemented using details of the architecture of the eyewear device **100**, as can other elements of network **995**.

(84) Memory **934** includes any storage device capable of storing various data and applications, including, among other things, color maps, camera data generated by the left and right visible light cameras **114**A-B and the image processor **912**, as well as images generated for display by the image display driver **942** on the see-through image displays **180**C-D of the optical assembly **180**A-B. While memory **934** is shown as integrated with high-speed circuitry **930**, in other examples, memory **934** may be an independent standalone element of the eyewear device **100**. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor **932** from the image processor **912** or low-power processor **922** to the memory **934**. In other examples, the high-speed processor **932** may manage addressing of memory **934** such that the low-power processor **922** will boot the high-speed processor **932** any time that a read or write operation involving memory **934** is needed.

(85) Server system **998** may be one or more computing devices as part of a service or network

computing system, for example, that include a processor, a memory, and network communication interface to communicate over the network **995** with the mobile device **990** and eyewear device **100**. Eyewear device **100** is connected with a host computer. For example, the eyewear device **100** is paired with the mobile device **990** via the high-speed wireless connection **937** or connected to the server system **998** via the network **995**.

(86) Output components of the eyewear device **100** include visual components, such as the left and right image displays **180**C-D of optical assembly **180**A-B as described in FIGS. **2**D-E (e.g., a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light emitting diode (LED) display, a projector, or a waveguide). The image displays **180**C-D of the optical assembly **180**A-B are driven by the image display driver **942**. The output components of the eyewear device **100** further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the eyewear device **100**, the mobile device **990**, and server system **998**, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

(87) Eyewear device **100** may optionally include additional peripheral device elements **919**. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with eyewear device **100**. For example, peripheral device elements **919** may include any I/O components including output components, motion components, position components, or any other such elements described herein.

(88) For example, the biometric components of the user interface field of view adjustment **900** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), WiFi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over wireless connections **925** and **937** from the mobile device **990** via the low-power wireless circuitry **924** or high-speed wireless circuitry **936**.

(89) According to some examples, an “application” or “applications” are program(s) that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, a third party application (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating systems. In this example, the third-party application can invoke API calls provided by the operating system to facilitate functionality described herein.

(90) FIG. **10** is a flowchart **1000** illustrating the operation of the eyewear device **100**/**200** and other components of the eyewear by the high-speed processor **932** executing instructions stored in memory, such as memory **934**. Although shown as occurring serially, the blocks of FIG. **10** may be

reordered or parallelized depending on the implementation.
(91) At block **1002**, the processor **932** instructs the array of microphones **145** to detect sounds in the environment, including sound from sound source **152**. The signals from the array of microphones are provided to the processor **932**.
(92) At block **1004**, the processor **932** processes the signals received by the array of microphones **145**. The processor **932** determines the direction of arrival of sound from the sound source **152** relative to the true forward-facing orientation of the eyewear device. The processor also identifies the sound source **152** by using the detection algorithm **945** and sound library **946**.
(93) At block **1006**, the processor **932** enables the respective light **262** in light array **260** as shown in FIG. **8**A-**8**B, the respective vibration device **272** in the vibration device array **270** as shown in FIG. **8**C-**8**D, and the indicator **275** in the image display **180**C-D as shown in FIG. **8**E-**8**F. As discussed, the light **262**, the vibration device **272**, and the indicator **275** that corresponds to the arrival direction of sound from the sound source **152** relative to the eyewear **100**/**200** is enabled. The relative brightness of the illuminated light **262** and the relative intensity of the vibration device **272** indicates the relative intensity of the sound from the sound source **152** relative the eyewear device **100**. The identity of the sound source **152** and the direction of sound arrival is communicated to the user by the displayed indicator **275**. The sound identity can also be indicated by the vibrational pattern of the vibration devices **272**. One or more sound sources **152** can be simultaneously tracked and multiple lights, vibration devices, and indicators can be simultaneously active if desired.
(94) FIG. **11** is a flowchart **1100** illustrating the operation of the eyewear device **100**/**200** and other components of the eyewear created by the high-speed processor **932** executing instructions stored in memory **934**. Although shown as occurring serially, the blocks of FIG. **11** may be reordered or parallelized depending on the implementation.
(95) At block **1102**, the processor **932** instructs the array of microphones **145** to detect sounds in the environment and the CV cameras **114**A-B to capture images and/or video of the environment.
(96) At block **1104**, the processor **932** processes the signals received by the array of microphones **145** and CV cameras **114**A-B. The processor **932** determines the direction of arrival of sound from the sound source **152** relative to the true forward-facing orientation of the eyewear device. The processor additionally identifies the sound source **152** by using the detection algorithm **945** and sound library **946**. The processor uses the CV camera to identify objects and correlates the identified sound source to the identified objects to assist in the identification of the identified sound source.
(97) At block **1106**, the processor **932** enables the respective light **262** in light array **260** as shown in FIG. **8**A-**8**B, the respective vibration device **272** in the vibration device array **270** as shown in FIG. **8**C-**8**D, and/or the indicator **275** in the image display **180**C-D as shown in FIG. **8**E-**8**F. As discussed, the light **262**, vibration device **272**, and indicator **275** that corresponds to the direction of the arriving sound from the sound source **152** relative to the eyewear device **100**/**200** is enabled. The relative brightness of the illuminated light **262** and the relative intensity of the vibration device **272** indicates the relative intensity of the sound from the sound source **152** relative the eyewear device **100**. The identity of the sound source **152** is communicated to the user by the indicator **275** displayed and/or by the vibrational pattern of the vibration devices **270**. In an example, a loud sound source **152** relative to the eyewear **100** is processed resulting in the display of an indictor **275**, the respective light **262** illuminating, and/or the respective vibration device **272** vibrating. One or more sound sources **152** can be simultaneously tracked and multiple lights, vibration devices, and indicators can be simultaneously active if desired.
(98) It will be understood that the terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein. Relational terms such as first and second and the like may be used solely to distinguish one entity or action

from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “includes,” “including,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises or includes a list of elements or steps does not include only those elements or steps but may include other elements or steps not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “a” or “an” does not, without further constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
(99) Unless otherwise stated, any and all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. Such amounts are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain. For example, unless expressly stated otherwise, a parameter value or the like may vary by as much as ±10% from the stated amount.
(100) In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, the subject matter to be protected lies in less than all features of any single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.
(101) While the foregoing has described what are considered to be the best mode and other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that they may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all modifications and variations that fall within the true scope of the present concepts.

Claims

1. Eyewear, comprising: a frame having a first side and a second side; a first temple adjacent the first side of the frame; an optical element supported by the frame; a computer vision (CV) camera supported by the frame and configured to perform object identification; a microphone array supported by the first temple and configured to receive a sound from a remote sound source that generates sound signals; a vibration device array configured to generate a vibration pattern indicative of the remote sound source; and an electronic processor configured to: process the sound signals received from the microphone array to determine a direction of the received sound relative to the eyewear; access a memory comprising a data set of known sounds correlated to known sound sources to identify the received sound; generate an indicator discernable by a user of the eyewear that is indicative of the direction of the received sound relative to the eyewear; and control the vibration device array to generate the vibration pattern indicative of the remote sound source, wherein a variable intensity of the vibration pattern is indicative of a variable intensity of the sound and varies directly with the intensity of the sound; wherein the processor is configured to identify the sound source using both the CV camera and the microphone array to identify the received sound by comparing the received sound to the memory data, and by correlating the received sound using the CV camera object identification.
2. The eyewear of claim 1, wherein the indicator indicates a direction of the received sound from the sound source that is outside a view of the optical element.
3. The eyewear of claim 1, wherein the optical element further comprises a see- through display configured to generate images, and the electronic processor is configured to display an image on

the display that is indicative of the sound source as the indicator.
4. The eyewear of claim 1, further comprising lights supported by the frame, wherein the lights comprise at least one left light positioned by the frame first side and at least one right light positioned by the frame second side, wherein the processor is configured to illuminate at least one of the lights as the indicator.
5. The eyewear of claim 4, wherein the lights comprise an array of lights that extend from the first side of the frame to the second side of the frame.
6. The eyewear of claim 5, wherein the lights are positioned on a bridge extending between the frame first side and the frame second side.
7. The eyewear of claim 6, wherein the processor is configured to illuminate one light of the array of lights at a brightness that corresponds to an intensity of a sound.
8. A method for use with eyewear, the eyewear having a frame, a first temple adjacent a first side of the frame, an optical element supported by the frame, a computer vision (CV) camera supported by the frame and configured to perform object identification, a microphone array supported by the first temple and configured to receive a sound from a remote sound source that generates sound signals, a vibration device array configured to generate a vibration pattern indicative of the remote sound source; and an electronic processor, the processor: processing the sound signals received from the microphone array to determine a direction of the received sound relative to the eyewear; accessing a memory comprising a data set of known sounds correlated to known sound sources to identify the received sound; generating an indicator discernable by a user of the eyewear that is indicative of the direction of the received sound relative to the eyewear; and controlling the vibration device array to generate the vibration pattern indicative of the remote sound source, wherein a variable intensity of the vibration pattern is indicative of a variable intensity of the sound and varies directly with the intensity of the sound; and identifying the sound source using both the CV camera and the microphone array to identify the received sound by comparing the received sound to the memory data, and by correlating the received sound using the CV camera object identification.
9. The method of claim 8, wherein the indicator indicates a direction of received sound from the sound source that is outside a view of the optical element.
10. The method of claim 8, wherein the optical element further comprises a see-through display configured to generate images, wherein the electronic processor is configured to display an image on the display that is indicative of the sound source as the indicator.
11. The method of claim 8, wherein the processor illuminates one light of the array of lights at a brightness that corresponds to an intensity of a sound.
12. A non-transitory computer-readable medium storing program code which, when executed, is operative to cause an electronic processor of eyewear having a frame, a first temple adjacent a first side of the frame, a computer vision (CV) camera supported by the frame and configured to perform object identification, an optical element supported by the frame, and a microphone array supported by the first temple and configured to receive a sound from a remote sound source that generates sound signals, a vibration device array configured to generate a vibration pattern indicative of the remote sound source; to perform the steps of: processing the sound signals received from the microphone array to determine a direction of the received sound relative to the eyewear; accessing a memory comprising a data set of known sounds correlated to known sound sources to identify the received sound; generating an indicator discernable by a user of the eyewear that is indicative of the direction of the received sound relative to the eyewear; controlling the vibration device array to generate the vibration pattern indicative of the remote sound source, wherein a variable intensity of the vibration pattern is indicative of a variable intensity of the sound and varies directly with the intensity of the sound; and identifying the sound source using both the CV camera and the microphone array to identify the received sound by comparing the received sound to the memory data, and by correlating the received sound using the CV camera object identification.

13. The non-transitory computer-readable medium as specified in claim 12, wherein the variable intensity of the vibration pattern varies directly with the intensity of the sound.